

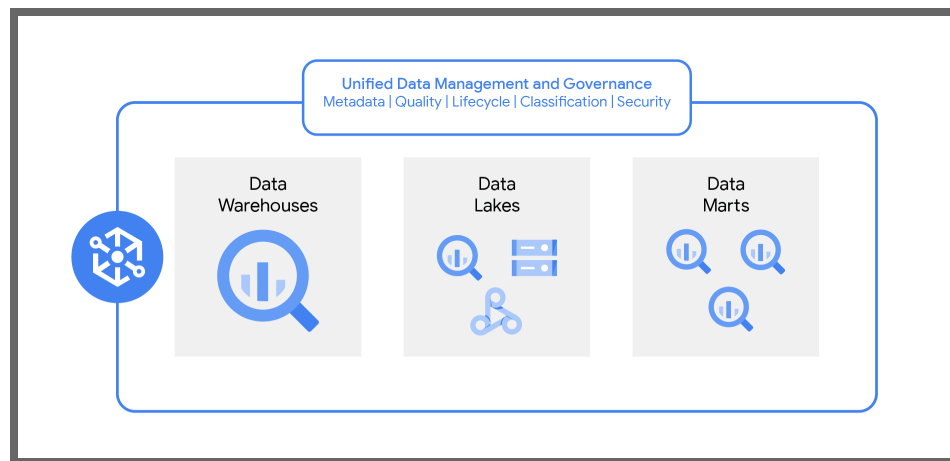
Overview of Dataplex

Previously, you learned that Dataplex is a data fabric that offers unified search and data discovery, all based on business context. Dataplex provides a centralized hub where users can explore, catalog, and manage data lakes, data warehouses, and data marts. In this reading, you'll learn more about Dataplex, its architecture, and how you can use it to access data.

The Dataplex architecture

As a data fabric, Dataplex provides centralized data access and security controls. This means that all the data in an organization falls under the same governance and rules. Dataplex brings together various data sources and services.

For example, consider an organization that has data warehouses, data lakes, and data marts. Dataplex provides management and governance, including metadata and security for all of the data in the organization.



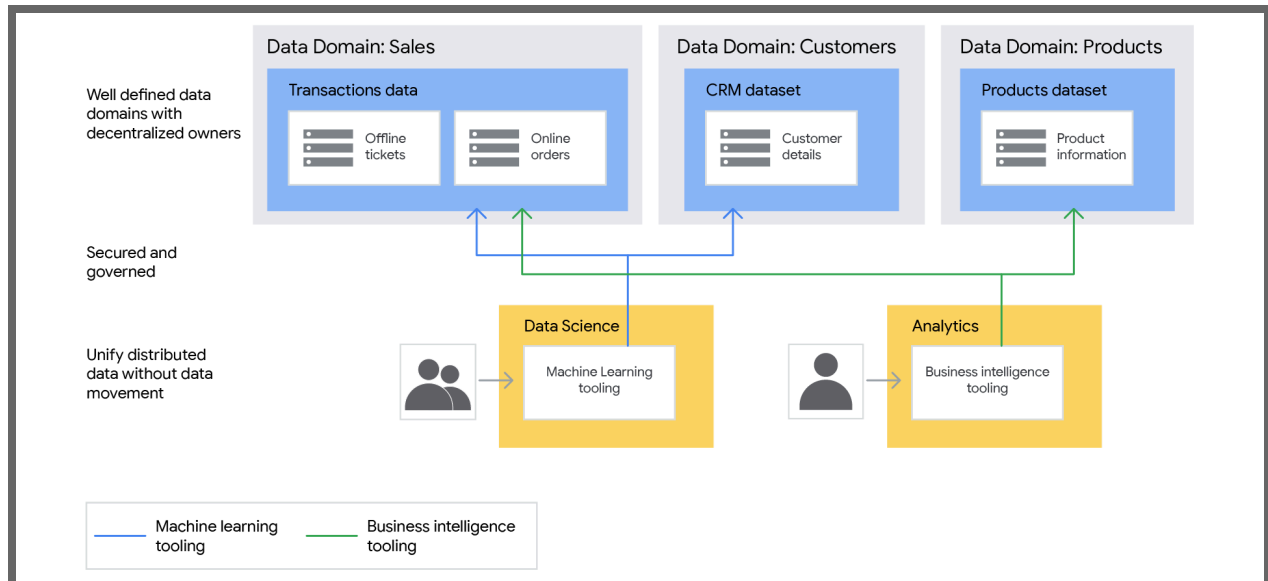
Elements of Dataplex

When using Dataplex, there are four components that build the data structure:

1. Lake: The logical domain that represents a business unit
2. Zone: A subdomain in the lake
3. Asset: Data that is mapped to storage
4. Entity: Represents metadata for structured, semi-structured, and unstructured data

The architecture of your existing data will influence the way Dataplex works, and the zones and lakes you are able to create.

Consider a company with three data domains. Each domain is in a Dataplex lake. Each portion of the organization or group controls the development, curation, and access to their own data lake. On the other end, data consumers request access to lakes and zones. The entire architecture is under the governance of Dataplex, making it easy to apply permissions and roles to all of the data in the organization.



Key takeaways

Dataplex is a data fabric that allows organizations to create management and governance rules for all the data held within the organization. It allows organizations to divide data into domains that include data lakes, data warehouses, and data marts, which provides users with access to all applicable data in a domain.

Resources for more information

To learn more about Dataplex and how it can be used to manage data organization-wide, check out these links:

- Infoworld provides an overview of Dataplex features: <https://www.infoworld.com/article/3692892/preview-google-cloud-dataplex-wows.html>
- Google Cloud provides information about Dataplex and how to implement it in your organization: <https://cloud.google.com/dataplex/docs/introduction#>